

GUANARE, Venezuela

WMO: 804280

Lat: 9.027N

Lon: 69.755W

Elev: 606

StdP: 14.38

Time Zone: -4.00 (W04)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	68.2	69.7	57.2	71.5	84.5	59.3	77.0	84.3	14.7	78.9	11.9	78.7	1.8	110	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.1	96.0	75.8	94.8	75.3	93.4	75.1	80.6	89.3	79.6	88.4	78.9	87.8	5.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	151.6	85.2	77.2	144.8	83.6	76.6	141.7	82.9	44.7	89.3	43.7	88.8	42.8	88.1	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	11.0	9.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	81.1	82.8	84.3	83.0	81.5	80.0	79.4	80.5	81.0	81.3	81.5	81.1
	DBStd	2.55	1.87	2.07	2.72	2.89	2.41	2.13	1.93	2.21	1.93	2.11	2.02	1.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11480	963	919	1063	991	976	901	911	944	931	971	945	965
	CDD65	6005	498	499	598	541	511	451	446	479	481	506	495	500
	CDH74	60906	5541	5829	7392	6112	5027	3836	3479	4244	4418	4934	4988	5106
	CDH80	26182	2656	2853	3765	2692	1899	1278	1193	1622	1789	2059	2142	2234
(14) Wind	WSAvg	3.9	4.3	4.8	4.8	4.2	3.9	3.8	3.5	3.6	3.4	3.5	3.6	3.9
(15) Precipitation	PrecAvg	55.2	0.9	0.4	0.4	3.3	6.3	11.1	8.9	7.0	5.5	5.7	4.1	1.3
	PrecMax	70.3	4.1	2.5	2.4	10.4	12.0	16.6	15.4	14.5	7.1	12.4	10.6	4.2
	PrecMin	41.9	0.0	0.0	0.0	0.0	2.9	5.9	0.7	4.0	3.5	0.0	0.5	0.0
	PrecStd	8.8	1.0	0.7	0.6	2.6	2.7	3.3	3.3	2.8	1.2	2.9	2.8	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.7	96.8	98.3	97.1	94.7	91.5	91.4	94.6	93.2	93.4	93.4	93.6
		MCWB	74.3	74.9	77.4	75.4	77.3	77.4	77.3	79.6	78.7	77.9	76.7	75.4
	2%	DB	93.1	94.8	96.6	95.2	91.8	89.8	89.6	91.3	91.6	91.7	91.8	92.7
		MCWB	73.8	74.4	76.4	75.2	76.8	76.9	76.7	77.9	77.9	77.5	76.4	75.1
	5%	DB	91.7	93.3	95.0	93.4	90.1	88.2	88.1	89.7	90.1	91.1	91.3	91.4
		MCWB	73.4	73.8	75.3	75.0	76.4	76.5	76.2	77.0	77.2	77.3	76.3	75.1
	10%	DB	90.9	91.8	93.4	91.4	88.9	87.2	86.7	88.0	89.3	89.6	89.8	89.8
		MCWB	73.2	73.5	74.6	74.7	76.1	76.2	76.0	76.4	76.9	76.8	76.1	74.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	78.7	82.8	80.3	80.8	80.5	79.8	81.2	80.7	80.9	80.0	79.1
		MCDB	90.7	91.4	95.0	88.5	88.6	87.4	86.6	88.6	89.8	89.5	89.0	88.5
	2%	WB	76.6	77.0	79.7	78.7	79.1	78.8	78.4	79.6	79.4	79.3	78.7	77.9
		MCDB	89.9	91.2	91.8	88.1	87.7	86.1	85.8	87.4	88.8	88.2	88.4	88.3
	5%	WB	75.4	75.8	77.6	77.7	78.1	77.8	77.5	78.4	78.4	78.3	77.8	76.9
		MCDB	88.4	89.7	90.2	86.8	86.6	85.1	84.9	86.3	87.5	87.2	87.6	87.4
	10%	WB	74.4	74.7	76.2	76.8	77.3	77.0	76.8	77.6	77.7	77.7	77.0	76.1
		MCDB	87.2	88.4	89.0	85.7	85.5	84.2	84.0	85.3	86.4	86.4	86.9	86.3
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	19.8	19.1	15.6	14.6	13.9	14.7	15.5	16.4	16.7	17.2	18.2
		MCDBR	22.3	21.5	20.7	19.1	16.6	16.1	17.2	17.8	18.1	18.4	19.1	20.3
	5% WB	MCWBR	8.2	7.8	7.7	6.7	7.2	7.6	7.7	7.4	7.6	7.6	7.1	7.4
		MCWBR	20.2	19.2	18.9	15.0	15.0	14.5	15.3	16.2	16.8	16.9	17.1	17.5
(40) Clear-Sky Solar Irradiance	taub		0.402	0.439	0.481	0.577	0.490	0.503	0.470	0.442	0.436	0.421	0.409	0.411
		taud	2.437	2.315	2.205	1.916	2.204	2.159	2.277	2.357	2.360	2.435	2.454	2.437
	Ebn at Noon		288	280	269	241	258	251	261	272	277	282	284	283
		Edh at Noon	36	43	48	63	46	48	43	40	40	37	36	36
(44) All-Sky Solar Radiation	RadAvg		1679	1778	1640	1424	1428	1422	1484	1555	1611	1566	1497	1558
	RadStd		84	80	163	120	93	75	84	70	71	53	72	100

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page